

Cipher To Silicon: The History of Computing Through Cryptography

Group 9's Members:

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Group Topic Theme: **Historical computing**

Throughout history, people have looked for ways to secure sensitive information. Early methods of encryption began with classical ciphers such as the Caesar cipher. However, as communication became more important, encryption methods also became more advanced. The codebreaking methods used before became time-consuming and inefficient as these systems became harder and more complicated to decipher. This led mathematicians and engineers to develop automated and mechanical solutions for encryption and codebreaking. In the early twentieth century, the development of polyalphabetic ciphers eventually contributed to the creation of electromechanical rotor machines. This includes the Hebern Rotor Machine and Arthur Scherbius's Enigma Machine.

This virtual exhibit project mainly focuses on cryptography and its relationship with the development of computing technologies. The increasing complexity of cryptographic systems before and during World War II made encrypted communications more difficult to decipher. This led to the development of specialized machines such as the Bombe and Colossus. By tracing the history of cryptography from classical ciphers to the Digital and Network Age, the project will explore how the growing challenges of encryption and codebreaking led to the development of specialized machines. These innovations eventually evolved into early general-purpose computers like the Harvard Mark I, ENIAC, and UNIVAC, helping shape modern computing. The exhibit will also highlight how these developments laid the foundation for modern encryption systems and internet security technologies that continue to protect information today.

The proposed topic will include the following interactive elements:

- **User Simulation** - The users will be able to simulate the operation of a Rotor Encryption module visually demonstrating how mechanical rotors laid the groundwork for early computing. Users interact with a simplified three-rotor model, customizing numeric offsets to observe how minor configuration shifts drastically alter encrypted outputs. Ultimately, this hands-on simulation highlights how the mathematical complexity of early cryptography necessitated the creation of programmable codebreaking machines.

- **Interactive Historical Timeline** - To guide users through the evolution of secure communications, the project utilizes a vertically scrolling historical timeline. This design choice allows for chronological tracking from c. 50 BCE through to the modern Internet Security Age. The timeline categorizes over 2,000 years of history into four distinct, filterable eras:
 - Classical: Highlighting algorithmic transformations like the Caesar Cipher.
 - Mechanical: Covering the shift to polyalphabetic and rotor-based automation.
 - Wartime Computing: Focusing on how cryptanalysis at Bletchley Park forced the invention of early electronic computers.
 - Digital & Network Age: Showcasing how cryptography became the foundation for modern web trust and personal encryption.
- **Codebreaking Challenge** - Users will be presented with encrypted messages and clues, allowing them to experience the challenges faced by wartime cryptanalysts and gain a better understanding of cryptographic principles. Using the knowledge gained, the users must attempt to decipher the messages within a limited number of attempts, gradually increasing in difficulty, introducing different concepts the higher the difficulty.

all are subject to change until the final submission.

Guide Snapshot

Project Info

- **Title:** Breaking Enigma: Secrets, War, and the Birth of Computing
- **Theme:** Historical computing
- **Members:** (Dustin Josh Bandigan, Nathan James Catapang, Raeka Estrella Hernaez, Paul Rhazzel Serrano, Chelsea Nichole Tamayo)

Color Palette

Aiming for an old war type vibe, Monochrome/Muted colors

- **Background:** #F5F5F5 (White-ish Gray)
- **Primary Text/UI:** #1A1A1A (Near Black)
- **Accents & Buttons:** #8E8E8E (Medium Gray)
- **Subtle UI Elements:** #D4D4D4 (Light Gray)
- **Cipher Output Text:** #4A4A4A (Dark Gray)
- **Others:** Any Color within the Grayscale spectrum

Typography

- **Headings & Titles:** *Special Elite or Courier Prime or Any Alternative*
- **Body Text:** *Inter or Open Sans*
- **Cipher/Encrypted Data:** *Fira Code or Monospace*

UI & Interactive Guidelines

- **User Simulation:** Mechanical keystroke animations with glowing paths tracing the signal through the simplified rotor-based cipher system.
- **Rotor Settings:** Coded to allow for modifying of rotor positions.
- **Timeline Layout:** Horizontal path with clickable circular nodes for exploring key events.
- **Timeline Pop-ups:** Clean modal cards displaying historical images, diagrams, and notable figures.
- **Codebreaking Challenge:** Clues styled as classified manila dossiers.
- **Codebreaking Inputs:** Terminal-style text fields for attempting to decipher the messages.
- **Difficulty Scaling:** Subtle visual static added to the background as cryptographic concepts increase in difficulty.

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VISUAL REPRESENTATION/S (NON-FINAL)

CIPHER TO SILICON

The History of Computing Through Cryptography

[Rotor Encryption](#)[Historical Timeline](#)[Practice Challenge](#)

Rotor Encryption

Explore one key chapter in the history of computing: rotor-based encryption. This interactive model shows how cryptography pushed engineers and mathematicians toward automated, programmable machines.

Why Rotors Matter In Computing History

A rotor is a mechanical letter-mapping wheel. Several rotors chained together create many possible substitutions, making encryption far harder to crack.

Historically, the need to break rotor ciphers drove large-scale codebreaking systems, which helped shape ideas used in early computing machines.

In this simplified model, each position (0-25) shifts the mapping by alphabet offset. Changing one rotor can completely change the output.

Rotor I00

Rotor II00

Rotor III00

Rotor I0

Rotor II0

Rotor III0

Enter Message

Type your message here ...

Encrypt

Reset

Encrypted Output

OUTPUT APPEARS HERE...

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Group 9 | Historical Computing and Cryptography Virtual Exhibit

Rotor Encryption Area

Rotor Encryption

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Rotor I

03

Rotor II

06

Rotor III

09

Rotor I

Rotor II

Rotor III

Encrypted Output

ZWDDG OGJDV!

Enter Message

H → [Rotor I:3] → [Rotor II:6] → [Rotor III:9] → Z

E → [Rotor I:3] → [Rotor II:6] → [Rotor III:9] → W

L → [Rotor I:3] → [Rotor II:6] → [Rotor III:9] → D

Encrypt

Reset

Rotor Encryption Area (When Encrypting a Message)

Shows how many places the letter/s are moved using the interactable rotors, shows how each letter was moved. Input of “HELLO WORLD!” encrypted output of “ZWDDG OGJDV!”

CIPHER TO SILICON

The History of Computing Through Cryptography

Rotor Encryption

Historical Timeline

Practice Challenge

Interactive Historical Timeline

Follow how cryptography evolved from manual ciphers to modern public-key systems, and how each leap influenced the development of computing.

Classical

Mechanical

Wartime Computing

Digital & Network Age

c. 50 BCE

Caesar Cipher

One of the earliest shift ciphers demonstrates algorithmic message transformation.

c. 850

Al-Kindi And Frequency Analysis

Systematic analysis introduces data-driven codebreaking, a precursor to computational thinking.

1467

Polyalphabetic Cipher Concepts

Leone Alberti proposes rotating alphabets, increasing complexity beyond single-shift systems.

1586

Vigenere Cipher Popularized

Polyalphabetic encryption becomes a major milestone in manual cryptography.

1854

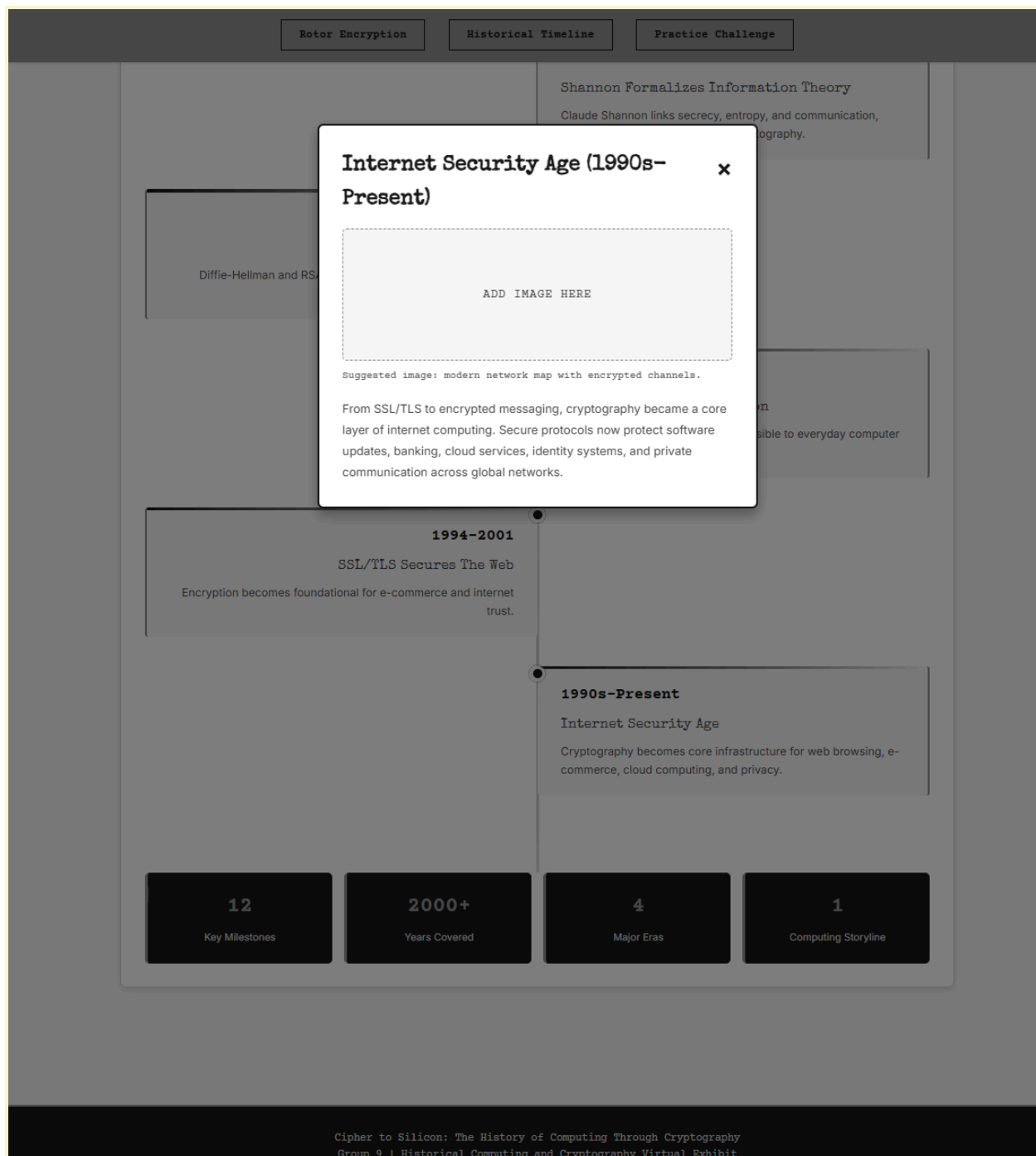
Babbage Breaks Vigenere

Analytical methods reveal hidden patterns, linking cryptanalysis and early computational reasoning.

Interactive Historical Timeline (START)



Interactive Historical Timeline (END)



Interactive Historical Timeline (When the Timeline is interacted with)

CIPHER TO SILICON

The History of Computing Through Cryptography

Rotor Encryption

Historical Timeline

Practice Challenge

Cryptography Practice Challenge

Practice classic cipher methods that represent different stages of cryptography history, from simple substitution to structured encoding.

Challenge Instructions

Select a difficulty level, decode each message, and connect each method to its historical era. You have limited attempts per case, and the harder modes reduce trial-and-error room.

Easy

Medium

Hard

CASE #1

CLASSIFIED

Clue: Simple substitution cipher. Each letter shifts by 3 positions.

How it works: Move each encrypted letter 3 steps backward in the alphabet.

KHOOR ZRUOG

Enter decrypted message ...

Submit Answer

CASE #2

CLASSIFIED

Clue: Caesar cipher with shift of 5.

How it works: Move each encrypted letter 5 steps backward in the alphabet.

MJQQT BTWQI

Enter decrypted message ...

Submit Answer

CASE #3

CLASSIFIED

Clue: ROT13 cipher - rotate each letter by 13 positions.

How it works: ROT13 swaps letters halfway across the alphabet (A=N, B=O).

RAVTZN EBPXF

Enter decrypted message ...

Submit Answer

CASE #4

CLASSIFIED

Clue: Atbash cipher - A=Z, B=Y, C=X, etc. (Reverse alphabet)

How it works: Replace each letter with its opposite partner in the alphabet.

YOVGXSOVB KZIP

Enter decrypted message ...

Submit Answer

Practice Challenges Tab

CLASSIFIED

CASE #1

Clue: Simple substitution cipher. Each letter shifts by 3 positions.

How it works: Move each encrypted letter 3 steps backward in the alphabet.

KHOOR ZRUOG

HELLO WORLD

Submit Answer

✓ CORRECT! You successfully decrypted the message.

CLASSIFIED

CASE #2

Clue: Caesar cipher with shift of 5.

How it works: Move each encrypted letter 5 steps backward in the alphabet.

MJQQT BTWQI

HI

Submit Answer

✗ FAILED. The answer was: HELLO WORLD

Practice Challenges Tab (Right and Wrong answer)

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